A Machine Learning-based Model for the Detection of Skin Cancer Using YOLOv10

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***Abstract*—Melanoma, the most dangerous form of skin cancer originates in pigment cells (melanocytes) and poses a significant threat to patients. Therefore, individuals must have any suspicious lesions or moles promptly evaluated for early detection to improve survival rates. The following study presents a Machine learning-based model for detecting skin cancer using YOLOv10 technology. The process involves preprocessing, augmentation, and training on two datasets: the ISIC 2017 dataset (with classes melanoma, nevus, and seborrheic keratosis) annotated based on YOLOv8 versions via Roboflow; and another dataset from Roboflow containing three classes—malignant tumors, benign growths, and healthy skin. Preprocessing techniques were utilized alongside augmentation methods to enhance the model's performance before fine-tuning the pre-trained YOLOv10 model as per requirements. This approach yielded promising results with achieved precision of 1, recall of 0.88, and mAP50 of 0.94 on the ISIC2017 dataset; precision of 1, recall of 0.95, mAP50 of 0.79, and F-score score of 82% on the Skin Detection Dataset respectively.**

Keywords*— *Skin lesion, Melanoma detection, localization, YOLOv10, annotation, augmentation

I. INTRODUCTION

Melanoma is a form of skin cancer that develops from melanocytes, the cells responsible for producing melanin pigment [1]. The melanocytes[2] make melanin in the skin which gives color to the skin. Skin cancer has two main types: melanoma and non-melanoma [3]. Melanoma often develops on sun-exposed skin areas. Annually, the WHO reports over 132,000 cases of melanoma and 3 million cases of non-melanoma skin cancer globally [4]. Over the past decade, the annual number of newly diagnosed cases of invasive melanoma increased by 32 percent [5]. The precise reason behind all melanomas is not fully understood. The majority of melanomas result from exposure to ultraviolet light. Early detection is crucial for effective treatment of melanoma. Successful treatment is possible when the condition is identified early. Melanoma, a type of skin cancer, can be fatal but is treatable if detected early[6]. Dermoscopic images are magnified views of the skin dermatologists use to examine and analyze skin lesions using methods such as ABCDE rules[7], the 7-Points checklist, CASH criteria, and the Menzies method [1]. Deep Learning is a subset of machine learning rapidly used in the health sector for diagnosis. Deep learning is capable of achieving favorable outcomes in the field of object detection[8]. Deep learning has achieved Significant object detection improvements, especially with CNNs. Object detection involves identifying objects in an image and indicating their location with a bounding box [9]. This object detection enables the identification and positioning of objects, making it easier to count and track them accurately with precise labeling. Object detection has many benefits in diagnosing skin cancer [10]. Advancements have been made in utilizing deep learning detection methods for identifying skin cancer. However, the latest and most advanced object recognition algorithms, such as YOLOv10, still need to be used. Our model's steps are depicted in Fig.1. The paper is organized into several sections. The first section is the introduction, followed by a review of relevant prior research. The third section describes the methodology, and the fourth section discusses the results. The paper concludes with a final section.

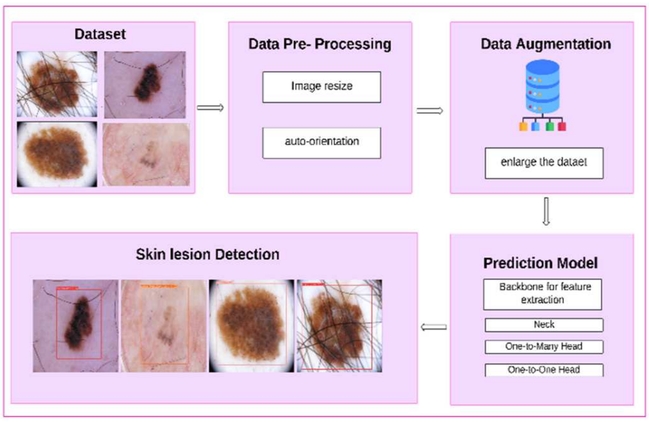

Fig. 1. Steps for Detection Model

II. RELATED WORK

Early prediction of skin cancer has been approached using various detection-based methods. This study introduces a novel method for improving the detection and segmentation of melanoma skin lesions. The approach was evaluated using the ISIC 2018 dataset, with the fine-tuned YOLOv3 model achieving a mean Average Precision of 96%, and the fine-tuned F-SegNet model outperforming state-of-the-art methods with an accuracy of 95.16%. Future research could further investigate using newer YOLO versions to enhance the performance [11]. This research focuses on skin lesion detection using pre-trained YOLO v3 and V4. The dataset contains 4389 images with 9 distinct classes, achieving a precision of 98.06% and a recall of 92.75%. The newer versions of the YOLO series introduce significantly faster performance than earlier ones [12]. M. Khan [13] proposed a multimodal fusion framework based on CNN for skin lesion segmentation and classification using five skin benchmark datasets. The model demonstrated a recall rate of 86.98% and an accuracy of 83.02% when evaluated on the HAM10000 dataset. However, these results are not sufficient. The study proposes using transfer learning and data augmentation techniques to enhance the accuracy of skin lesion detection with YOLOv7 on-edge devices, resulting in an 82.1% detection rate for skin lesions. However, it is necessary to utilize newer versions of YOLO [14]. This research[15] examined the performance of four integrated networks (from YOLOv3 to YOLOv7) in categorizing skin lesions. The models were trained using a standard dataset and evaluated based on lesion localization, accuracy of classification, and time taken for inference. Notably, YOLOv7 exhibited outstanding results with an IoU of 86.3%, mAP of 75.4%, F1-measure of 80%, and an inference time of 0.32 seconds per image. Nonetheless, further enhancements could be made by leveraging the latest Yolo model available.

III. MATERIALS AND METHODS

A. Dataset

The YOLOv10 model requires a dataset configuration approach in YOLO format. This approach outlines the main directory of the dataset, relative paths to directories containing training/validation/testing images, *.txt files containing annotations, and a list of class names.

The 2017 ISIC [16] challenge dataset was released in 2017. The dataset contains 2750 JPEG images, each measuring 767 × 102 pixels. The images are categorized into three specific groups: benign nevi, seborrheic keratosis, and melanoma.

The SKIN_CANCER_DETECTION V2[17] dataset from Roboflow is used in this study, specifically intended for skin detection. It comprises three classes: healthy, benign, and malignant.

B. Augmentation

The images have been annotated in Roboflow for detection-based techniques. The ISIC 2017 dataset has been extended to 546 images by applying different mentioned techniques[18]. The image has been resized to 640 x 640 and an auto-orientation method has been employed during the preprocessing stage. Three augmented outputs have been generated for each training example. This included 90° rotations (clockwise and counter-clockwise), hue adjustments between -15° and +15°, and saturation variations between -31% and +31%. These changes have been made to increase model accuracy.

The size of the SKIN_CANCER_DETECTION V2 dataset has been expanded to 1197 images. These images were resized to 800 × 800 using an automated orientation technique in the preprocessing stage. Furthermore, horizontal flips and brightness shifts of -45% to +45% were added during augmentation.

In Table I, we provide a detailed summary of the original and augmented datasets, highlighting the differences between the training and validation sets for both datasets.The Detection Model

The latest model in the YOLO[19] series, YOLOv10 [9], is an advanced version that incorporates object detection for computer vision—developed by Multimedia Intelligence Group, School of Software, and Tsinghua University. YOLOv10 has undergone extensive testing on standard benchmarks such as COCO, showcasing superior performance and efficiency.YOLOv10 comprises two primary elements: 1. Non-Maximum Suppression-eliminated training and inference, and 2. streamlined and precise model formulation. YOLOv8 consists of five distinct variations - Nano, small, medium, balanced, and large - each customized to meet specific needs. The Nano version is intended for resource-constrained environments, while the large version is designed to achieve maximum accuracy [20]. The YOLOv10 model demonstrates greater precision and speed than earlier versions of YOLO and other advanced models. For instance, the YOLOv10-S shows a 1.8 times speed improvement compared to RT-DETR-R18 while maintaining similar average precision on the COCO dataset. In addition, the YOLOv10-B demonstrates a 46% reduction in latency and uses 25% fewer parameters than the equivalent-performing YOLOv9-C [21]. We chose YOLOv10s because our dataset is relatively small, and this model offers six different variants designed for various sizes and efficiency levels.

C. Performance Evaluation

This section outlines the evaluation metrics used to assess the performance of the skin lesion detection model. The recall rate represents the proportion of true positive object detections among all actual objects in the image. Precision denotes the fraction of detected objects that are correctly identified, rather than being false positives[7]. The F1-score[22] synthesizes precision and recall to yield a single metric encapsulating a classification model's overall effectiveness[15]. Mean Average Precision (mAP) [15] is a summary metric aggregating precision and recall across

TABLE I. SUMMARY OF DATASET

Dataset	Original Dataset			Augmented Dataset		
	Training	**Validation set**	**Testing**	**Training**	**Validation set**	**Testing**
ISIC 2017	159	47	22	420	104	22
Skin Cancer Detection	627	---	---	855	230	112

multiple object classes, offering a comprehensive assessment of the model's performance in object detection tasks. The confusion matrix[23] is a fundamental tool used to evaluate the performance of a classification algorithm.

D. Training Setting

When training the YOLOv10 model, we applied transfer learning by utilizing the pre-trained weights of its smaller version, YOLOv10s, from the COCO dataset. We then set the hyperparameters for our datasets. We use the Google Colab T4 GPU for this experiment.

The configuration parameters for the YOLOv10 model can be found in Table II Results. The model demonstrated promising results on the dataset, exhibiting improved accuracy and efficiency. Table III summarizes the model's performance using this evaluation matrix.

Fig. 2 and 3 present the achieved prediction details. The confidence scores represent the model's certainty about its predictions, with higher scores indicating greater confidence in the classification. The results demonstrate the model's capability to classify skin lesions with varying confidence levels. High-confidence predictions suggest strong performance, while lower-confidence predictions identify areas for further improvement.

The confusion matrix depicted in Fig. 4 and 5 provides detailed insights into the model's performance. The confusion matrix of the ISIC Dataset suggests the model performs the most accurately in identifying nevus and seborrheic keratosis lesions. However, it struggles to distinguish between melanoma and background clearly and exhibits confusion between melanoma and nevus lesions. While Fig. 6 and 7 show the overall performance of the model. The training losses exhibit a decreasing trend, indicating effective learning. The evaluation metrics display an increasing trend, reflecting improved performance on unseen data. The classification losses initially show a sharp decrease, followed by stabilization, suggesting a rapid initial learning phase. The recall and precision metrics demonstrate a good balance between identifying relevant instances and minimizing false positives. Overall, the model is training effectively and improving over time. Compared to recent studies, the detailed evaluation metrics for the proposed framework are presented in Table IV.

TABLE II. PARAMETERS DETAILS

Parameters	Value
image size	640
batch size	8
epochs	70

TABLE III. PERFORMANCE SUMMARY

Dataset	Precision	Recall	F1 score	mAP50
ISIC 2017	1	0.98	0.88	0.94
Skin Cancer Detection	1	0.95	0.79	0.82

TABLE IV. COMPARISON OF YOLOV10 PERFORMANCE WITH RELEVANT STATE-OF-THE-ART APPROACHES ON ISIC 2017

Method	Dataset	Precision	Recall	F1 score	mAP50
Yolov7 from [15]	ISIC 2017	--	--	79.6	78.0
Yolov8 from[24]	ISIC2018	0.93	0.92	--	0.973
Fine-Tuned YOLOv10	ISIC 2017	1	0.98	0.88	0.94

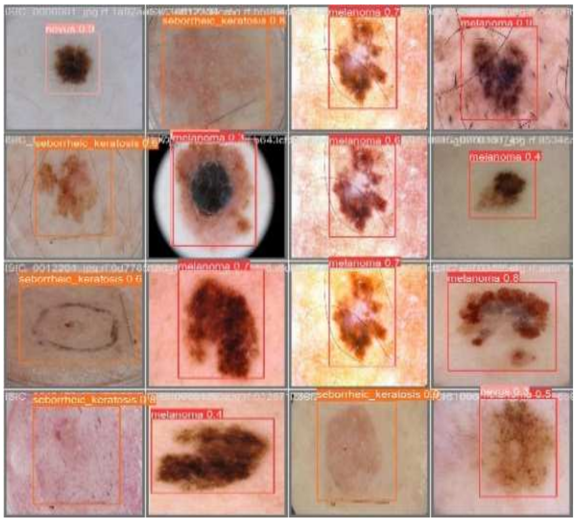

Fig. 2. Detection results of Yolov10 for dataset ISIC 2017

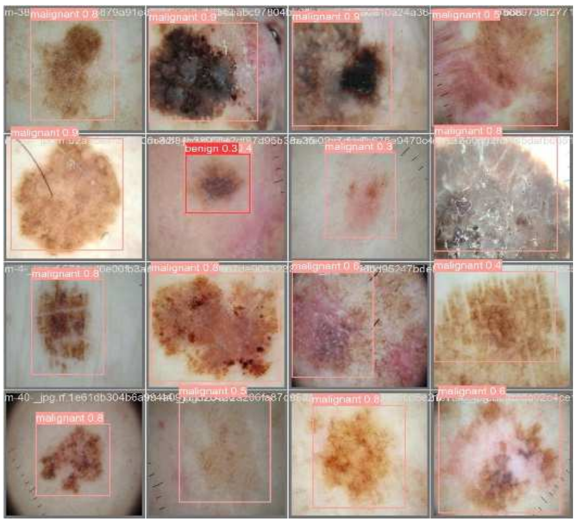

Fig. 3. Detection results of Yolov10 dataset Skin Cancer Detection

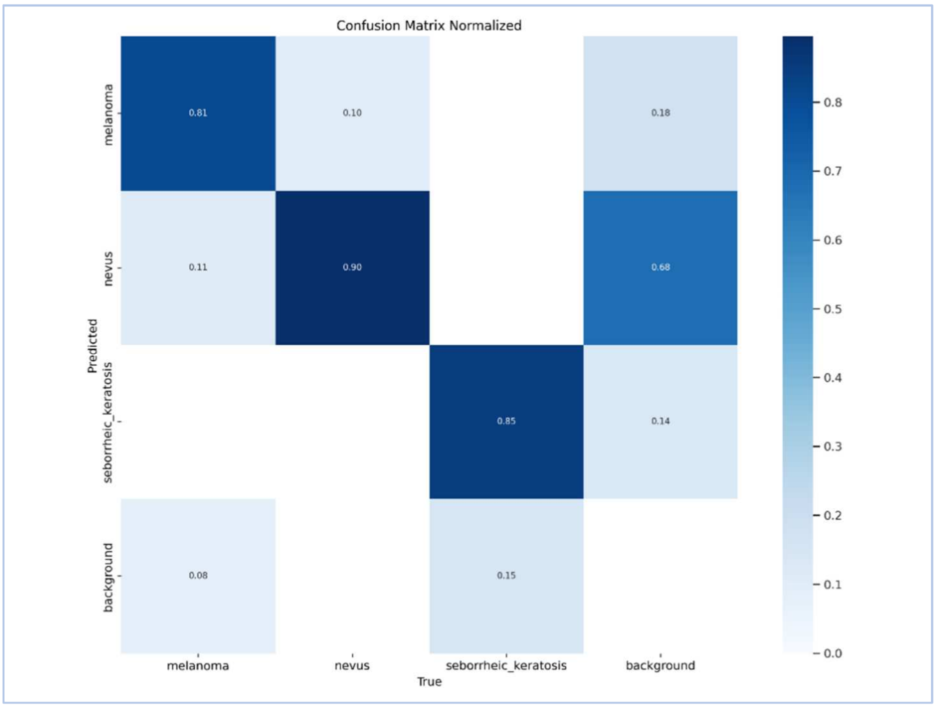

Fig. 4. Confusion matrix result for ISIC 2017 dataset

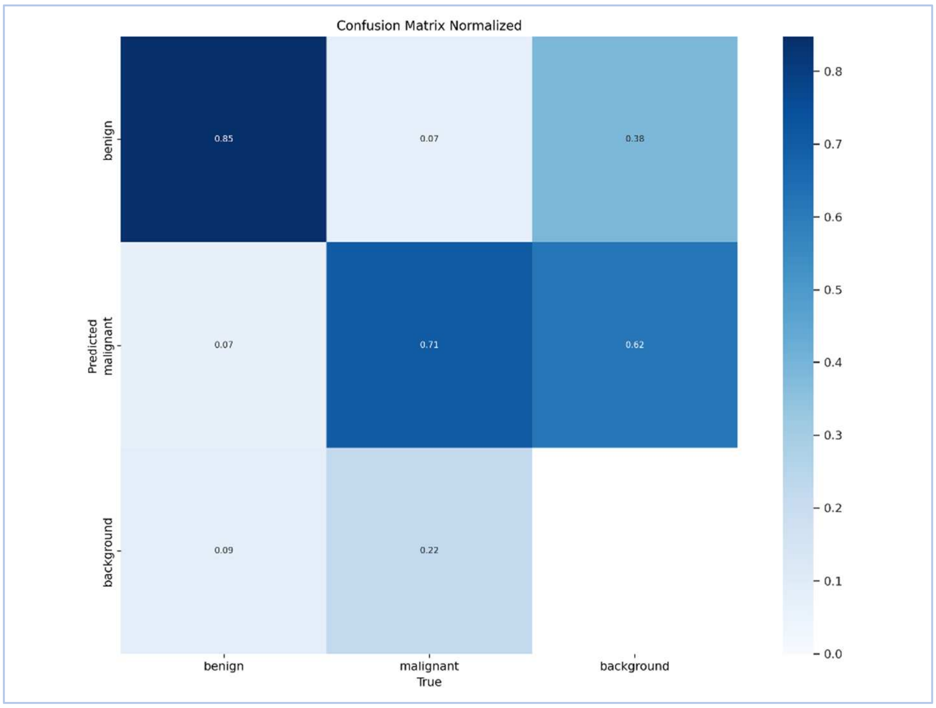

Fig. 5. Confusion matrix result for Skin cancer Detection dataset

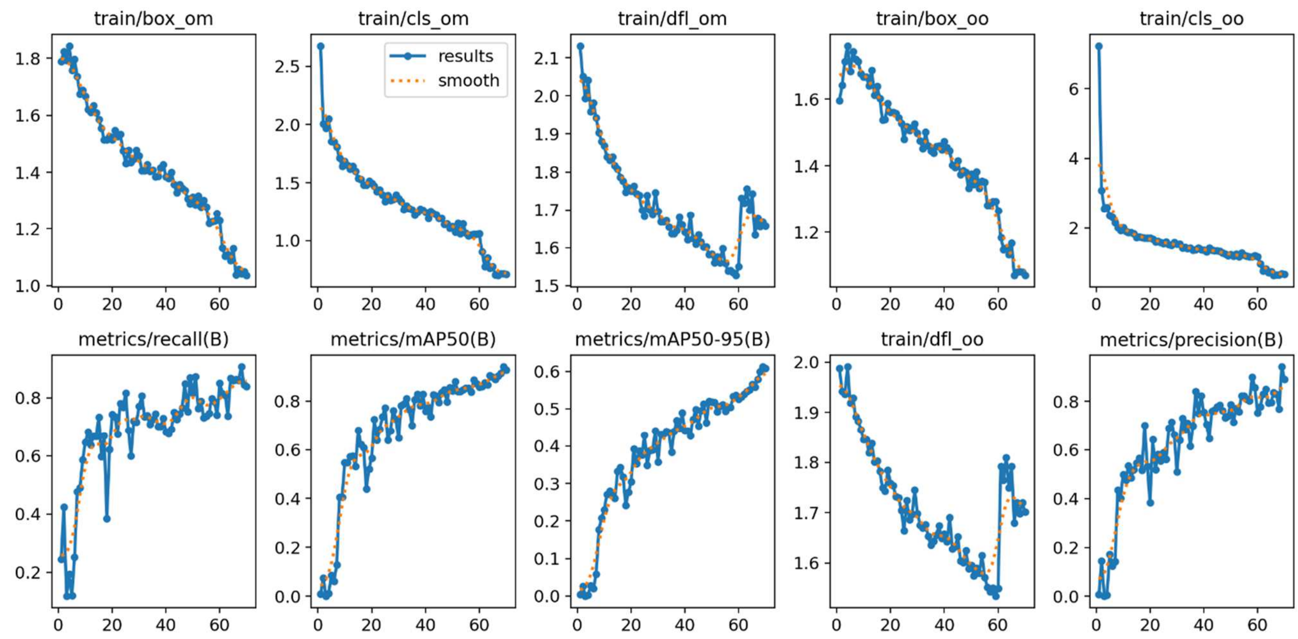

Fig. 6. Results Summary of Yolov10 for dataset ISIC 2017

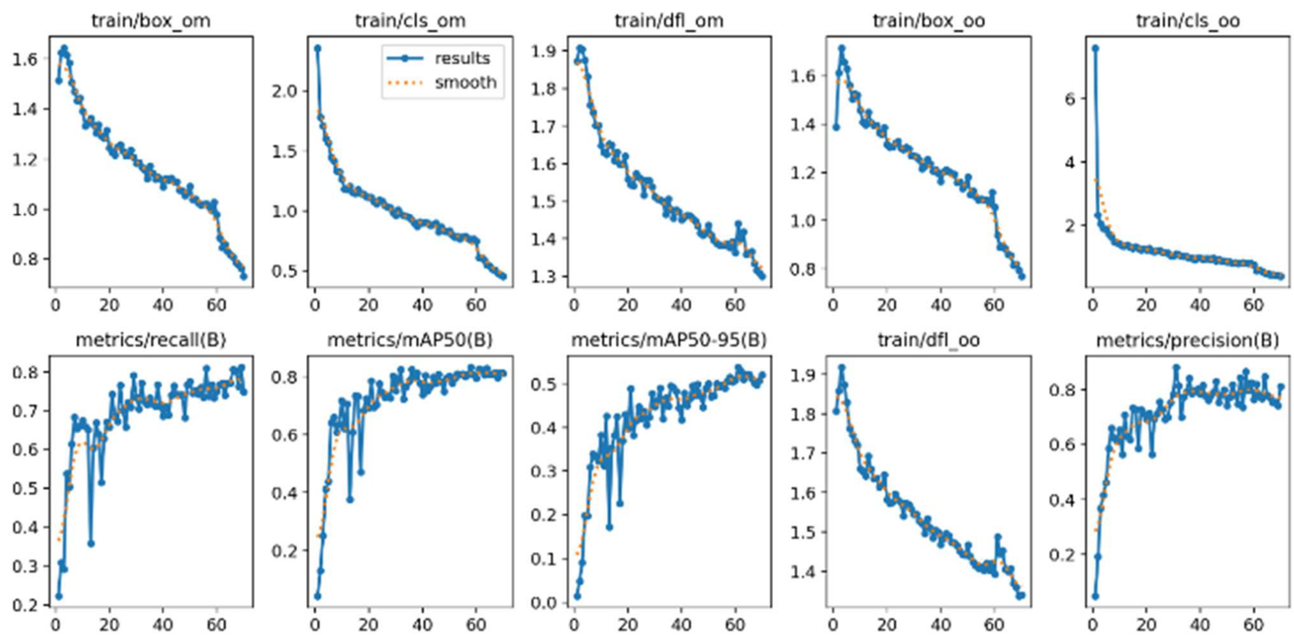

Fig. 7. Results Summary of Yolov10 for dataset Skin cancer Detection

IV. CONCLUSION

Fast and accurate detection of skin lesions is crucial for the healthcare industry. This study centers on utilizing the advanced YOLOv10 object detection model to create a highly effective and efficient system for detecting skin lesions. By leveraging the ISIC 2018 dataset, an AI object detection model was developed to precisely identify the presence and locations of skin lesions in dermoscopic images. The achieved precision, recall, and mAP50 are 1, 0.98, and 0.94 respectively.

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